

28.2 Restricted eigenvalue conditions

In Parts II and III, the precision of least squares was governed by the smallest eigenvalue of $\mathbf{X}^\top \mathbf{X}$: the worst variance of $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$ over unit vectors $\mathbf{a}$ is $\sigma^2 / \lambda_{\min}(\mathbf{X}^\top \mathbf{X})$ (Proposition 19.6.1). When $p > n$ that eigenvalue is zero. This section explains which weaker condition takes its place once $\boldsymbol{\beta}$ is assumed sparse, and shows with numbers how demanding it is.

28.2.1 Notation

Throughout Sections 28.2–28.5 the columns of $\mathbf{X}$ are scaled so that $\|\mathbf{x}_j\|^2 = n$ for every j , and $\hat{\boldsymbol{\Sigma}} = \mathbf{X}^\top \mathbf{X} / n$ is the sample Gram matrix, with unit diagonal. For a vector $\mathbf{v} \in \mathbb{R}^p$ and a set $S \subseteq \{1, \dots, p\}$, $\mathbf{v}_S$ is the vector that agrees with $\mathbf{v}$ on S and is zero off S , so that $\mathbf{v} = \mathbf{v}_S + \mathbf{v}_{S^c}$. The ℓ_1 and ℓ_∞ norms are $\|\mathbf{v}\|_1 = \sum_j |v_j|$ and $\|\mathbf{v}\|_\infty = \max_j |v_j|$, and $\|\mathbf{v}\|$ is still the Euclidean norm. The **support** of $\boldsymbol{\beta}$ is $S = \{j : \beta_j \neq 0\}$, and $s = |S|$. A vector with at most s nonzero entries is **s -sparse**. By Cauchy–Schwarz (Proposition 1.1.1), for every $\mathbf{v}$,

$$\|\mathbf{v}_S\|_1 \leq \sqrt{s} \|\mathbf{v}_S\| \leq \sqrt{s} \|\mathbf{v}\|. \quad (28.2.1)$$

28.2.2 Sparsity restores identifiability

If $\boldsymbol{\beta}$ is only known to lie in $\mathbb{R}^p$, then $\boldsymbol{\beta}$ and $\boldsymbol{\beta} + \mathbf{v}$ with $\mathbf{v} \in \mathcal{N}(\mathbf{X})$ give the same mean, and nothing can separate them. If $\boldsymbol{\beta}$ is known to be sparse, most of those alternatives are excluded.

Proposition 28.2.1. Identifiability of sparse vectors

Distinct s -sparse vectors $\boldsymbol{\beta} \neq \boldsymbol{\beta}'$ always have $\mathbf{X}\boldsymbol{\beta} \neq \mathbf{X}\boldsymbol{\beta}'$ iff every set of $2s$ columns of $\mathbf{X}$ is linearly independent. In particular this requires $2s \leq n$.

Proof. If $\boldsymbol{\beta}, \boldsymbol{\beta}'$ are s -sparse and $\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\beta}'$, then $\mathbf{v} = \boldsymbol{\beta} - \boldsymbol{\beta}'$ is a vector in $\mathcal{N}(\mathbf{X})$ with at most $2s$ nonzero entries. It is zero if every $2s$ columns are independent. Conversely, if some $2s$ columns are dependent, there is a nonzero $\mathbf{v} \in \mathcal{N}(\mathbf{X})$ supported on them. Split its support into two sets of at most s indices, T_1 and T_2 , and put $\boldsymbol{\beta} = \mathbf{v}_{T_1}$, $\boldsymbol{\beta}' = -\mathbf{v}_{T_2}$. These are distinct s -sparse vectors with $\mathbf{X}\boldsymbol{\beta} - \mathbf{X}\boldsymbol{\beta}' = \mathbf{X}\mathbf{v} = \mathbf{0}$. Finally, $2s$ vectors in $\mathbb{R}^n$ can be independent only if $2s \leq n$. ■

Identifiability is a statement about exactly sparse vectors and noiseless data. For estimation we need a quantitative version that allows for noise and for estimates that are only approximately sparse. The key observation, proved in [Section 28.3](#), is that the error of the lasso, $\boldsymbol{\Delta} = \hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}$, lies with high probability in a particular cone. Its ℓ_1 mass off the support is at most three times its mass on the support. Curvature of the least squares criterion is needed only in the directions of that cone.

28.2.3 Two conditions on a cone

Definition 28.2.1. Compatibility and restricted eigenvalue conditions

Let $S \subseteq \{1, \dots, p\}$ have $s \geq 1$ elements, and let

$$\mathcal{K}(S) = \{\mathbf{v} \in \mathbb{R}^p : \|\mathbf{v}_{S^c}\|_1 \leq 3\|\mathbf{v}_S\|_1\}. \quad (28.2.2)$$

The **compatibility constant** and the **restricted eigenvalue** of the design for S are

$$\phi^2(S) = \min_{\mathbf{v} \in \mathcal{K}(S), \mathbf{v} \neq \mathbf{0}} \frac{s \mathbf{v}^\top \hat{\Sigma} \mathbf{v}}{\|\mathbf{v}_S\|_1^2}, \quad \kappa^2(S) = \min_{\mathbf{v} \in \mathcal{K}(S), \mathbf{v} \neq \mathbf{0}} \frac{\mathbf{v}^\top \hat{\Sigma} \mathbf{v}}{\|\mathbf{v}\|^2}.$$

The design satisfies the **compatibility condition** for S if $\phi^2(S) > 0$, and the **restricted eigenvalue condition** for S if $\kappa^2(S) > 0$. The same definitions with a nonnegative definite matrix Σ in place of $\hat{\Sigma}$ give $\phi_\Sigma^2(S)$ and $\kappa_\Sigma^2(S)$.

A nonzero $\mathbf{v} \in \mathcal{K}(S)$ has $\mathbf{v}_S \neq \mathbf{0}$, since otherwise $\|\mathbf{v}_{S^c}\|_1 \leq 0$. So both ratios are defined. Both are unchanged when $\mathbf{v}$ is rescaled, and the sets $\{\mathbf{v} \in \mathcal{K}(S) : \|\mathbf{v}\| = 1\}$ and $\{\mathbf{v} \in \mathcal{K}(S) : \|\mathbf{v}_S\|_1 = 1\}$ are closed and bounded. So the minima are attained, which justifies writing min. The constant 3 comes from the choice of penalty in [Section 28.3](#). Other choices give other constants and the same theory.

The cone contains the s -dimensional subspace of vectors supported on S but no larger subspace ([Exercise \(C1\)](#)). So the null space of $\mathbf{X}$, of dimension at least $p - n$, can still avoid it.

Proposition 28.2.2. Basic facts

- (a) $\lambda_{\min}(\hat{\Sigma}) \leq \kappa^2(S) \leq \phi^2(S)$. If $p > n$, then $\lambda_{\min}(\hat{\Sigma}) = 0$.
- (b) $\kappa^2(S) > 0$ iff $\phi^2(S) > 0$ iff $\mathcal{N}(\mathbf{X}) \cap \mathcal{K}(S) = \{\mathbf{0}\}$.
- (c) (*Transfer.*) If Σ is symmetric and $\max_{j,k} |\hat{\Sigma}_{jk} - \Sigma_{jk}| \leq \delta$, then $\kappa^2(S) \geq \kappa_\Sigma^2(S) - 16s\delta$ and $\phi^2(S) \geq \phi_\Sigma^2(S) - 16s\delta$.
- (d) (*Coherence.*) If $\mu = \max_{j \neq k} |\hat{\Sigma}_{jk}|$ is the **mutual coherence** of the scaled design, then $\kappa^2(S) \geq 1 - 16s\mu$ and $\phi^2(S) \geq 1 - 16s\mu$.

Proof.

- (a) The first inequality holds because $\kappa^2(S)$ minimizes the Rayleigh quotient of $\hat{\Sigma}$ over a subset of nonzero vectors, while $\lambda_{\min}(\hat{\Sigma})$ minimizes it over all of them (Theorem 1.7.7). For the second, [Eq. 28.2.1](#) gives $\|\mathbf{v}_S\|_1^2 \leq s\|\mathbf{v}\|^2$, so $s \mathbf{v}^\top \hat{\Sigma} \mathbf{v} / \|\mathbf{v}_S\|_1^2 \geq \mathbf{v}^\top \hat{\Sigma} \mathbf{v} / \|\mathbf{v}\|^2 \geq \kappa^2(S)$ for every $\mathbf{v} \in \mathcal{K}(S)$. If $p > n$, then $\text{rank}(\hat{\Sigma}) \leq n < p$, so $\hat{\Sigma}$ is singular.
- (b) $\mathbf{v}^\top \hat{\Sigma} \mathbf{v} = \|\mathbf{X}\mathbf{v}\|^2/n$ vanishes iff $\mathbf{X}\mathbf{v} = \mathbf{0}$. Since the minima are attained, each constant is zero iff some nonzero $\mathbf{v} \in \mathcal{K}(S)$ has $\mathbf{X}\mathbf{v} = \mathbf{0}$.
- (c) For $\mathbf{v} \in \mathcal{K}(S)$, $\|\mathbf{v}\|_1 = \|\mathbf{v}_S\|_1 + \|\mathbf{v}_{S^c}\|_1 \leq 4\|\mathbf{v}_S\|_1$. Hence

$$|\mathbf{v}^\top (\hat{\Sigma} - \Sigma) \mathbf{v}| \leq \sum_{j,k} |v_j| |v_k| \delta = \delta \|\mathbf{v}\|_1^2 \leq 16\delta \|\mathbf{v}_S\|_1^2 \leq 16s\delta \|\mathbf{v}\|^2,$$

the last step by [Eq. 28.2.1](#). Dividing by $\|\mathbf{v}\|^2$ gives $\mathbf{v}^\top \hat{\Sigma} \mathbf{v} / \|\mathbf{v}\|^2 \geq \mathbf{v}^\top \Sigma \mathbf{v} / \|\mathbf{v}\|^2 - 16s\delta$. Dividing the middle bound by $\|\mathbf{v}_S\|_1^2/s$ gives $s \mathbf{v}^\top \hat{\Sigma} \mathbf{v} / \|\mathbf{v}_S\|_1^2 \geq \phi_\Sigma^2(S) - 16s\delta$.

- (d) Take $\Sigma = \mathbf{I}$ in (c), with $\delta = \mu$ (the diagonals agree). Then $\kappa_\Sigma^2(S) = 1$, and $\phi_\Sigma^2(S) \geq$

$$\kappa_{\mathbf{I}}^2(S) = 1 \text{ by (a) applied to } \mathbf{I}.$$

■

By (b) the two conditions agree about *whether* the constants vanish, but by (a) their sizes, which enter the bounds, can differ a great deal: a lower bound on $\kappa^2(S)$ is the stronger assumption. The fast prediction rate of [Section 28.3](#) uses $\phi^2(S)$, the ℓ_2 bound $\kappa^2(S)$. Part (c) carries a population condition to the sample when $\hat{\Sigma}$ is uniformly close to Σ , and (d) is the case $\Sigma = \mathbf{I}$.

28.2.4 When the conditions hold

Populations. If $\lambda_{\min}(\Sigma) > 0$, part (a) applied to Σ gives $\phi_{\Sigma}^2(S) \geq \kappa_{\Sigma}^2(S) \geq \lambda_{\min}(\Sigma)$. For example, the equicorrelation matrix $\Sigma = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$, $0 \leq \rho < 1$, has $\mathbf{v}^\top \Sigma \mathbf{v} = (1 - \rho)\|\mathbf{v}\|^2 + \rho(\mathbf{1}^\top \mathbf{v})^2 \geq (1 - \rho)\|\mathbf{v}\|^2$, so both constants are at least $1 - \rho$ for every p and S . The difficulty lies in going from Σ to $\hat{\Sigma}$ when $p > n$.

Random designs. Suppose the rows of $\mathbf{X}$ are independent $N_p(\mathbf{0}, \Sigma)$ vectors with unit variances. A union bound over the p^2 entries of $\hat{\Sigma} - \Sigma$, each an average of n products of normal variables, gives $\max_{jk} |\hat{\Sigma}_{jk} - \Sigma_{jk}| \leq C\sqrt{\log p/n}$ with high probability once $n \geq C'\log p$. The tail bound for each entry is not the normal bound of the next section, since products of normal variables have heavier tails; it is a Bernstein-type inequality, not proved here (see Wainwright 2019, chapter 2). Part (c) then gives a positive restricted eigenvalue once n exceeds a multiple of $s^2 \log p$. This is wasteful, because it bounds $\mathbf{v}^\top (\hat{\Sigma} - \Sigma) \mathbf{v}$ by the worst entry. Raskutti, Wainwright and Yu (2010) proved, by an argument we do not reproduce, that with probability tending to one

$$\frac{\|\mathbf{X}\mathbf{v}\|}{\sqrt{n}} \geq c_1 \|\Sigma^{1/2} \mathbf{v}\| - c_2 \sqrt{\frac{\log p}{n}} \|\mathbf{v}\|_1 \quad \text{for all } \mathbf{v} \in \mathbb{R}^p,$$

for constants $c_1, c_2 > 0$. On the cone, $\|\mathbf{v}\|_1 \leq 4\sqrt{s}\|\mathbf{v}\|$, so this gives $\kappa(S) \geq c_1 \lambda_{\min}(\Sigma)^{1/2} - 4c_2 \sqrt{s \log p/n}$, which is positive once n exceeds a multiple of $s \log p$, possibly far smaller than p .

28.2.5 When they fail

The conditions fail when a vector of the cone is (nearly) invisible to $\mathbf{X}$.

Duplicated columns. If $\mathbf{x}_k = \mathbf{x}_j$ with $j \in S$ and $k \notin S$, then $\mathbf{v} = \mathbf{e}_j - \mathbf{e}_k$ has $\|\mathbf{v}_{S^c}\|_1 = \|\mathbf{v}_S\|_1 = 1$, so it lies in the cone, and $\mathbf{X}\mathbf{v} = \mathbf{0}$. Both constants are zero. Nearly equal columns give nearly zero constants ([Exercise \(B1\)](#)). No estimator can say whether the effect belongs to $\mathbf{x}_j$ or to its copy.

Short combinations of inactive columns. More generally, suppose an active column is a combination of inactive ones, $\mathbf{x}_j = \mathbf{X}_{S^c} \mathbf{a}$ with $\|\mathbf{a}\|_1 \leq 3$. Then $\mathbf{v} = \mathbf{e}_j - \mathbf{a}$ (with $\mathbf{a}$ placed in the coordinates of S^c) lies in the cone and $\mathbf{X}\mathbf{v} = \mathbf{0}$. When $p > n$ the inactive columns usually span $\mathbb{R}^n$, so the conditions require every such combination to have a large ℓ_1 norm.

Too many active variables for the sample size. Even with independent regressors and a population compatibility constant of one, the sample constant can be close to zero if s is large relative to $n/\log p$. A combination of hundreds of inactive columns with total ℓ_1 weight 3 can then imitate $\mathbf{X}_S \mathbf{v}_S$ closely. The next example measures this.

Example. Compatibility constants of Gaussian designs

Take $p = 1000$ regressors, $S = \{1, \dots, s\}$, and three random designs, with columns scaled

so that $\|\mathbf{x}_j\|^2 = n$: independent standard normal entries with $n = 200$ (the design used in [Section 28.3](#)) and with $n = 400$, and equicorrelated normal regressors with $\rho = 0.5$ and $n = 400$. The compatibility constant is a minimum of a convex quadratic over the cone. For each sign pattern of $\mathbf{v}_S$ the problem is convex, and a duality gap certifies the computed value. The table gives $\phi^2(S)$ and the mutual coherence μ ; its rows are labelled by the common correlation ρ .

Design	μ	$s = 1$	$s = 2$	$s = 5$	Population
$\rho = 0$, $n = 200$	0.350	0.328	0.146	0.0045	1
$\rho = 0$, $n = 400$	0.238	0.479	0.384	0.177	1
$\rho = 0.5$, $n = 400$	0.641	0.266	0.192	0.089	≥ 0.5

The sample constants lie far below the population values, even for $s = 1$, and fall quickly with s . At $n = 200$ and $s = 5$ the constant 0.0045 makes the fast-rate bound of the next section numerically useless for that design; doubling n helps a great deal. The coherence bound of [Proposition 28.2.2\(d\)](#) is useless throughout: with $\mu = 0.350$, $1 - 16s\mu$ is -4.60 already at $s = 1$.

```
import numpy as np

def proj_simplex(u):
    """Euclidean projection onto {u >= 0, sum(u) = 1}."""
    v = np.sort(u)[::-1]
    css = np.cumsum(v) - 1
    k = np.flatnonzero(v - css / np.arange(1, len(u) + 1) > 0)[-1]
    return np.maximum(u - css[k] / (k + 1), 0)

def proj_l1(w, radius):
    """Euclidean projection onto the l1 ball of the given radius."""
    if np.abs(w).sum() <= radius:
        return w
    return np.sign(w) * proj_simplex(np.abs(w) / radius) * radius

def compatibility(X, S, radius=3.0, iters=4000):
    """Return  $\phi^2(S)$  and a certified lower bound (value minus duality gap)."""
    n, p = X.shape
    s = len(S)
    Sc = np.setdiff1d(np.arange(p), S)
    A, C = X[:, S], X[:, Sc]
    L = 2 * s / n * np.linalg.norm(X, 2) ** 2          # Lipschitz constant of the
                                                         # gradient
    best, lower = np.inf, np.inf
    for code in range(2 ** (s - 1)):                    # sign patterns of  $\mathbf{v}_S$  up to a
                                                         # global sign
        sg = np.array([1.0] + [-1.0 if (code >> k) & 1 else 1.0 for k in range(s -
1)])
        u, w = np.full(s, 1 / s), np.zeros(len(Sc))
        yu, yw, t = u.copy(), w.copy(), 1.0
        for _ in range(iters):                          # accelerated projected
                                                         # gradient
            r = A @ (sg * yu) + C @ yw
```

```

    gu, gw = 2 * s / n * sg * (A.T @ r), 2 * s / n * (C.T @ r)
    u_new, w_new = proj_simplex(yu - gu / L), proj_l1(yw - gw / L, radius)
    t_new = (1 + np.sqrt(1 + 4 * t * t)) / 2
    yu = u_new + (t - 1) / t_new * (u_new - u)
    yw = w_new + (t - 1) / t_new * (w_new - w)
    u, w, t = u_new, w_new, t_new
    r = A @ (sg * u) + C @ w
    value = s / n * r @ r
    gu, gw = 2 * s / n * sg * (A.T @ r), 2 * s / n * (C.T @ r)
    gap = gu @ u - gu.min() + gw @ w + radius * np.abs(gw).max()
    best, lower = min(best, value), min(lower, value - gap)
    return best, lower

rng = np.random.default_rng(2820)
n, p = 100, 300
X = rng.normal(size=(n, p))
X /= np.sqrt(np.sum(X ** 2, axis=0) / n) # ||x_j||_2 = n
print("smallest eigenvalue of X^T X / n:", np.linalg.eigvalsh(X.T @ X / n)[0])
for s in [1, 2, 3]:
    value, lower = compatibility(X, np.arange(s), iters=1500)
    print(f"s = {s}: phi^2 = {value:.3f}")

```

[Run this code online](#)

Remark (Checking the conditions in practice). The constants depend on the unknown S and cannot be checked from data. They describe the designs for which the theory is informative: those in which no short combination of columns imitates another. With groups of nearly equivalent regressors, common in genomics, the lasso may still predict well (by the slow rate of [Section 28.3](#)), but its choice among them is arbitrary.

28.2.6 Exercises

28.2.6.1 A. Check your understanding

Exercise. A1

Let $p = 2$ and $S = \{1\}$. Sketch $\mathcal{K}(S)$ and show that it is not convex. Show that for general S it is the union of 2^s convex cones, one for each sign pattern of $\mathbf{v}_S$. This is why the example above solves one convex problem per sign pattern.

28.2.6.2 B. Practice

Exercise. B1

Let $j \in S$ and $k \notin S$. Show that $\phi^2(S) \leq s\|\mathbf{x}_j - \mathbf{x}_k\|^2/n$ and $\kappa^2(S) \leq \|\mathbf{x}_j - \mathbf{x}_k\|^2/(2n)$. If $\mathbf{x}_j^\top \mathbf{x}_k/n = r$ (the cosine of the angle between the two columns, which is their sample correlation when the columns are centred), conclude that $\phi^2(S) \leq 2s(1 - r)$.

Solution. Take $\mathbf{v} = \mathbf{e}_j - \mathbf{e}_k \in \mathcal{K}(S)$. Then $\|\mathbf{v}_S\|_1 = 1$, $\|\mathbf{v}\|^2 = 2$ and $\mathbf{v}^\top \hat{\Sigma} \mathbf{v} = \|\mathbf{x}_j - \mathbf{x}_k\|^2/n$, which gives both bounds. With $\|\mathbf{x}_j\|^2 = \|\mathbf{x}_k\|^2 = n$, $\|\mathbf{x}_j - \mathbf{x}_k\|^2/n = 2 - 2\mathbf{x}_j^\top \mathbf{x}_k/n = 2(1 - r)$.

Exercise. B2

For $\Sigma = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$ with $0 \leq \rho < 1$ and $p \geq 2$, show that $\kappa_\Sigma^2(S) = 1 - \rho$ for every S . Is the equicorrelation structure a hard case for the restricted eigenvalue condition?

Solution. The lower bound $1 - \rho$ was shown in the text. For equality, it is enough to find a nonzero $\mathbf{v} \in \mathcal{K}(S)$ with $\mathbf{1}^\top \mathbf{v} = 0$, since then $\mathbf{v}^\top \Sigma \mathbf{v} = (1 - \rho)\|\mathbf{v}\|^2$. If $s \geq 2$, take $\mathbf{e}_j - \mathbf{e}_k$ with $j, k \in S$. If $s = 1$, take $\mathbf{e}_j - \mathbf{e}_k$ with $S = \{j\}$ and any $k \neq j$; it lies in the cone because $1 \leq 3$. The population condition is therefore as good as the smallest eigenvalue, and equicorrelation is not a hard case in the population. It is harder in the sample (the table above), and harder still for exact support recovery, as [Section 28.4](#) shows for a related design.

28.2.6.3 C. Going deeper**Exercise. C1**

Show that every linear subspace contained in $\mathcal{K}(S)$ has dimension at most s , and that the bound is attained. Deduce that if $\dim \mathcal{N}(\mathbf{X}) > p - s$ then the conditions fail for every S of size s , whereas $\dim \mathcal{N}(\mathbf{X}) = p - n$ with $n \geq s$ is not excluded by this dimension count.

Exercise. C2

(Exact recovery by ℓ_1 minimization.) Say $\mathbf{X}$ has the **null space property** for S if every nonzero $\mathbf{v} \in \mathcal{N}(\mathbf{X})$ has $\|\mathbf{v}_{S^c}\|_1 > \|\mathbf{v}_S\|_1$. Show that this holds iff, for every $\boldsymbol{\beta}$ supported on S , $\boldsymbol{\beta}$ is the unique solution of: minimize $\|\mathbf{b}\|_1$ subject to $\mathbf{X}\mathbf{b} = \mathbf{X}\boldsymbol{\beta}$. Relate the property to [Proposition 28.2.2\(b\)](#) with the constant 3 replaced by 1.

Solution. Suppose the property holds, $\boldsymbol{\beta}$ is supported on S , and $\mathbf{X}\mathbf{b} = \mathbf{X}\boldsymbol{\beta}$ with $\mathbf{b} \neq \boldsymbol{\beta}$. Then $\mathbf{v} = \mathbf{b} - \boldsymbol{\beta} \in \mathcal{N}(\mathbf{X})$ is nonzero, and $\mathbf{b}_S = \boldsymbol{\beta} + \mathbf{v}_S$, $\mathbf{b}_{S^c} = \mathbf{v}_{S^c}$, so $\|\mathbf{b}\|_1 \geq \|\boldsymbol{\beta}\|_1 - \|\mathbf{v}_S\|_1 + \|\mathbf{v}_{S^c}\|_1 > \|\boldsymbol{\beta}\|_1$. So $\boldsymbol{\beta}$ is the unique minimizer. Conversely, if some nonzero $\mathbf{v} \in \mathcal{N}(\mathbf{X})$ has $\|\mathbf{v}_{S^c}\|_1 \leq \|\mathbf{v}_S\|_1$, put $\boldsymbol{\beta} = \mathbf{v}_S$ and $\mathbf{b} = -\mathbf{v}_{S^c}$. Then $\mathbf{X}\mathbf{b} = \mathbf{X}\boldsymbol{\beta} - \mathbf{X}\mathbf{v} = \mathbf{X}\boldsymbol{\beta}$, $\mathbf{b} \neq \boldsymbol{\beta}$ and $\|\mathbf{b}\|_1 \leq \|\boldsymbol{\beta}\|_1$, so $\boldsymbol{\beta}$ is not the unique minimizer. The property says exactly that $\mathcal{N}(\mathbf{X})$ meets the cone $\{\|\mathbf{v}_{S^c}\|_1 \leq \|\mathbf{v}_S\|_1\}$ only at zero. This is part (b) with the constant 1 in place of 3. The noisy problem needs the larger cone because the penalty must also absorb the noise.